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November 4, 2025, 2:54 PM

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CAS Lexicon Search:

Mukaiyama aldol reaction - Preferred Concept

Search Tasks

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Exported: Retrieved Related Reaction Results + Filters (2,596)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	

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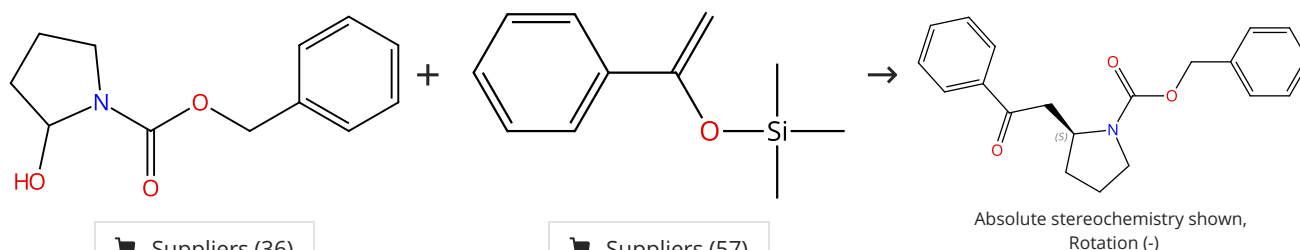
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (36)

 Suppliers (57)

31-614-CAS-31113843

Steps: 1 Yield: 100%

Catalytic Asymmetric Additions of Enol Silanes to In Situ Generated Cyclic, Aliphatic N-Acyliminium Ions

By: Grossmann, Oleg; et al

Angewandte Chemie, International Edition (2022), 61(9), e202115036.

1.1 Catalysts: 2759168-43-7

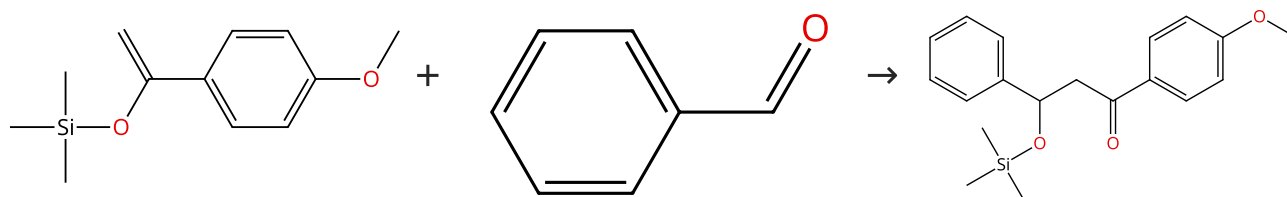
Solvents: Diethyl ether; 16 h, -70 °C

1.2 Reagents: Triethylamine, Water; 10 min, -70 °C; -70 °C → rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (6)

 Suppliers (69)

31-330-CAS-13037866

Steps: 1 Yield: 100%

Sc^{III}-Doped Zeolites as New Heterogeneous Catalysts: Mukaiyama Aldol Reaction

By: Olmos, Andrea; et al

Chemistry - A European Journal (2009), 15(42), 11229-11234, S11229/1-S11229/11.

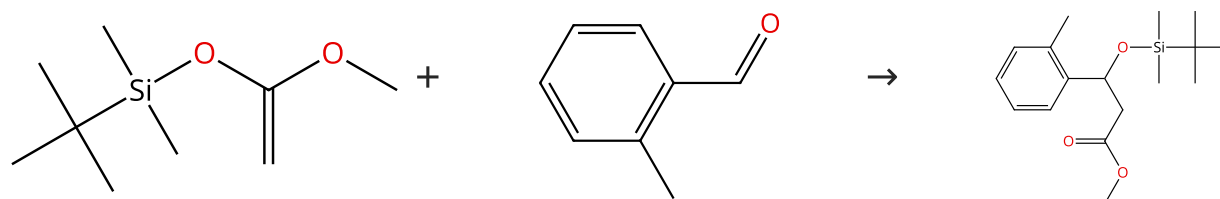
1.1 Catalysts: Scandium triflate

Solvents: Dichloromethane; 15 min, 20 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (55)

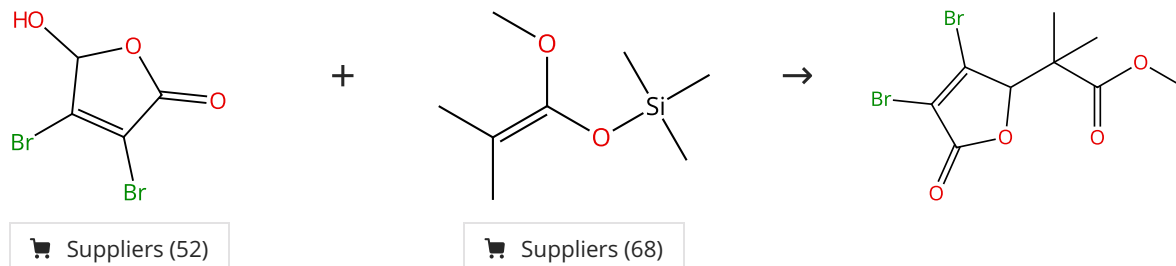
 Suppliers (87)

 Supplier (1)

31-330-CAS-12069530 Steps: 1 Yield: 100% 1.1 Catalysts: Silver hexafluoroantimonate, Dichloro[<i>N,N'</i>-[(2,6-pyridinediyl-κ<i>N</i>)diethyldiyn]bis[2,6-diethylbenzenamine-κ<i>N</i>]] iron Solvents: Dichloromethane; 1 h, rt 1.2 Solvents: Dichloromethane; overnight, rt Experimental Protocols	New bis(imino)pyridine complexes of iron(II) and iron(III), and their catalytic activity in the Mukaiyama aldol reaction By: Shejwalkar, Pushkar; et al Synthesis (2014), 46(1), 57-66, 10 pp..
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Scheme 4 (1 Reaction)

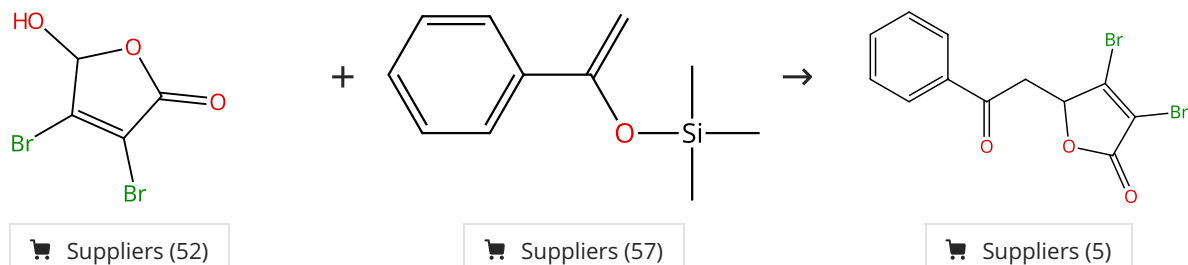
Steps: 1 Yield: 100%



31-330-CAS-5594866 Steps: 1 Yield: 100% 1.1 Catalysts: Zinc chloride Solvents: Toluene; rt → -20 °C 1.2 2 h, -20 °C; 3 h, -20 °C → 18 °C; 16 h, 18 °C	Mucohalic acid in Lewis acid catalyzed Mukaiyama aldol reaction: a concise method for highly functionalized γ-substituted γ-butenolides By: Angell, Paul; et al Tetrahedron Letters (2005), 46(12), 2029-2032.
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Scheme 5 (1 Reaction)

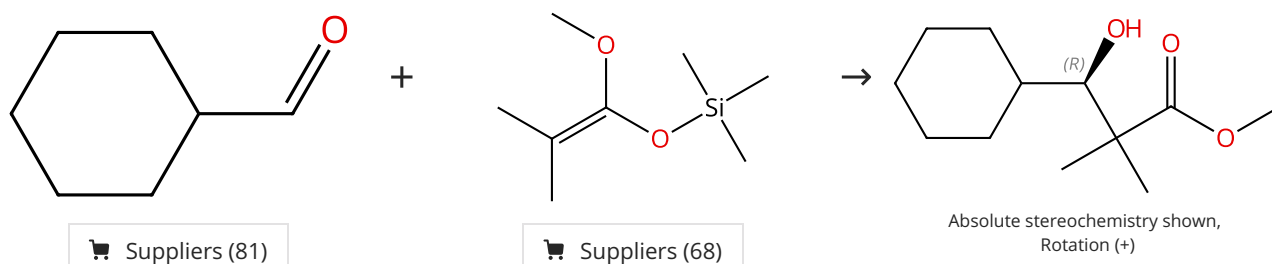
Steps: 1 Yield: 100%



31-330-CAS-2872909 Steps: 1 Yield: 100% 1.1 Catalysts: Scandium triflate Solvents: Diethyl ether; rt → -20 °C 1.2 2 h, -20 °C; 3 h, -20 °C → 18 °C; 16 h, 18 °C	Mucohalic acid in Lewis acid catalyzed Mukaiyama aldol reaction: a concise method for highly functionalized γ-substituted γ-butenolides By: Angell, Paul; et al Tetrahedron Letters (2005), 46(12), 2029-2032.
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Scheme 6 (1 Reaction)

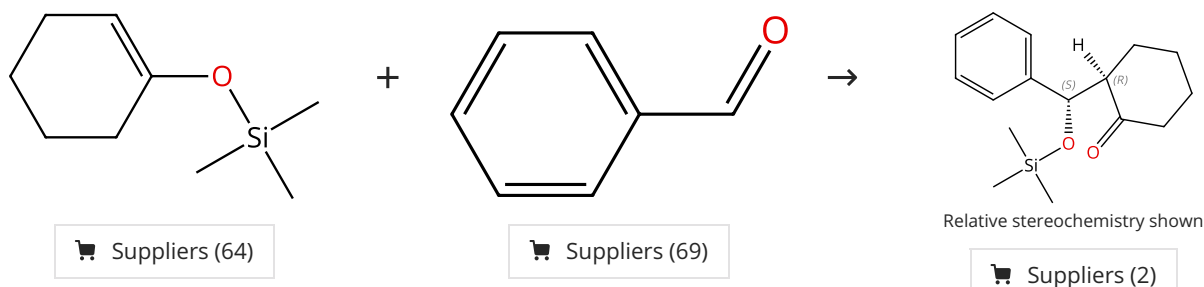
Steps: 1 Yield: 100%



31-330-CAS-6890410 Steps: 1 Yield: 100% 1.1 Catalysts: Silver, (acetato-$\kappa O, \kappa O'$)[(1<i>S</i>)-[1,1'-binaphthalene]-2,2'-diylbis[diphenylphosphine-κP]]-, (7<i>S</i>)- Solvents: Dimethylformamide 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water 1.4 Solvents: Diethyl ether	Mechanism of silver(I)-catalyzed Mukaiyama aldol reaction: active species in solution in AgPF₆-(<i>S</i>)-BINAP versus AgOAc-(<i>S</i>)-BINAP systems By: Ohkouchi, M.; et al Journal of Molecular Catalysis A: Chemical (2001), 170(1-2), 1-15.
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Scheme 7 (1 Reaction)

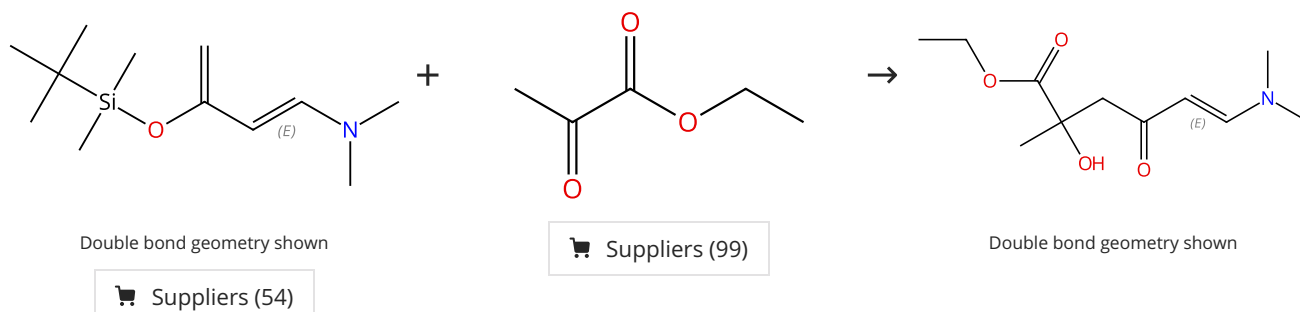
Steps: 1 Yield: 100%



31-330-CAS-12056552 Steps: 1 Yield: 100% 1.1 Solvents: Dichloromethane; 15 min, rt	Efficient catalysis of mesoporous Al-MCM-41 for Mukaiyama aldol reactions By: Iwanami, Katsuyuki; et al Catalysis Communications (2009), 10(15), 1990-1994.
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Scheme 8 (1 Reaction)

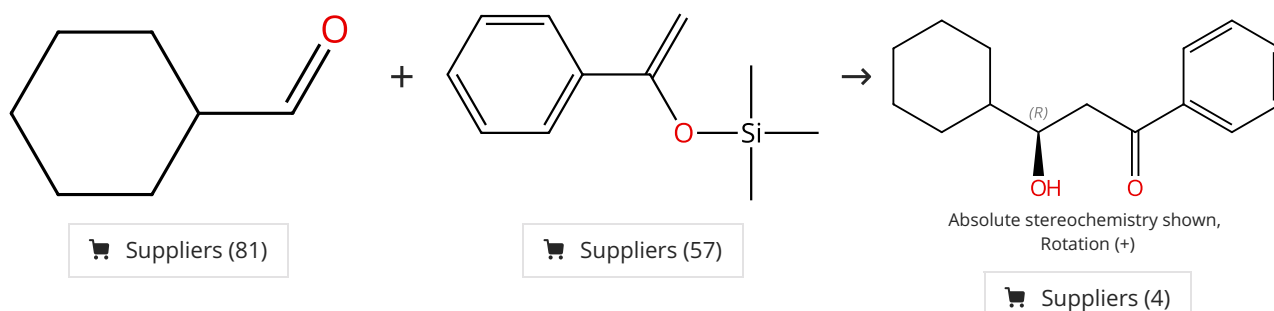
Steps: 1 Yield: 100%



31-330-CAS-4189603 Steps: 1 Yield: 100% 1.1 Solvents: Water; 1 h, rt Experimental Protocols	Water opportunities: catalyst and solvent in Mukaiyama aldol addition of Rawal's diene to carbonyl derivatives By: De Rosa, Margherita; et al Tetrahedron (2011), 67(33), 5949-5955.
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Scheme 9 (1 Reaction)

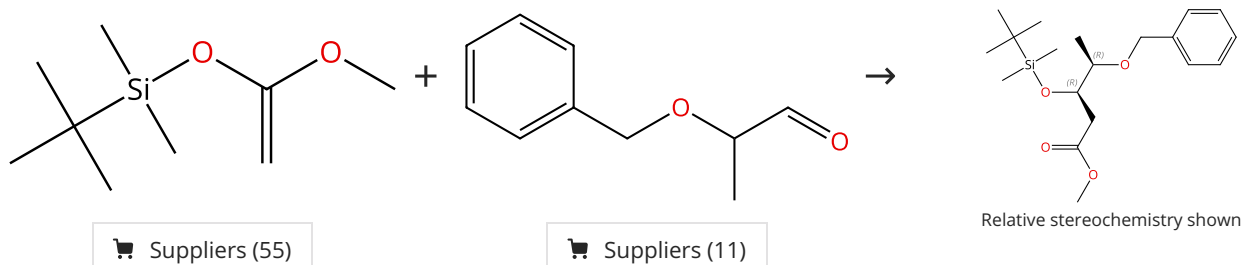
Steps: 1 Yield: 100%



31-330-CAS-12448314 Steps: 1 Yield: 100% 1.1 Catalysts: Silver acetate, (-)-BINAP Solvents: Dimethylformamide, Water 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water	Silver(I)-catalyzed asymmetric Mukaiyama aldol reaction By: Ohkouchi, Munetaka; et al Enantiomer (2000), 5(1), 71-81.
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Scheme 10 (1 Reaction)

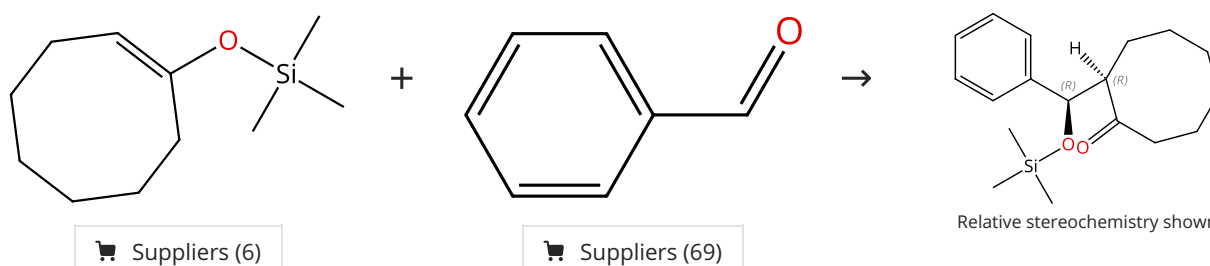
Steps: 1 Yield: 100%



31-330-CAS-11766453 Steps: 1 Yield: 100% 1.1 Reagents: Lithium perchlorate Solvents: Diethyl ether	A rapid injection NMR study of the chelation controlled Mukaiyama aldol addition: TiCl_4 versus LiClO_4 as the Lewis acid By: Reetz, Manfred T.; et al Tetrahedron (1992), 48(27), 5731-42.
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Scheme 11 (1 Reaction)

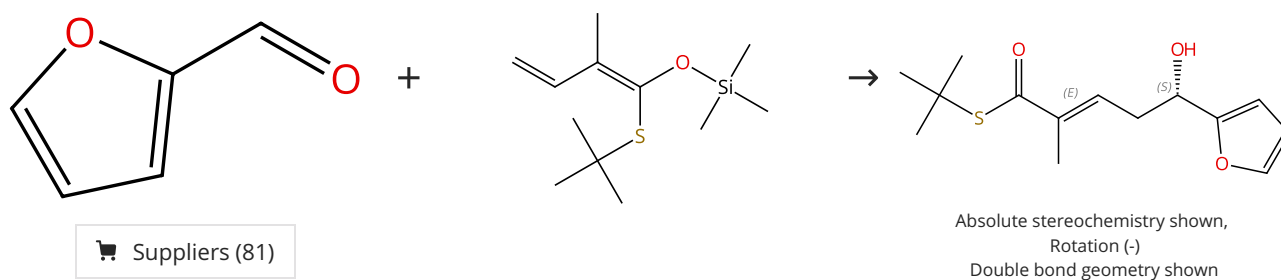
Steps: 1 Yield: 100%



31-330-CAS-10622786 Steps: 1 Yield: 100% 1.1 Catalysts: Scandium triflate Solvents: Dichloromethane; 15 h, 20 °C Experimental Protocols	Sc^{III}-Doped Zeolites as New Heterogeneous Catalysts: Mukaiyama Aldol Reaction By: Olmos, Andrea; et al Chemistry - A European Journal (2009), 15(42), 11229-11234, S11229/1-S11229/11.
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Scheme 12 (1 Reaction)

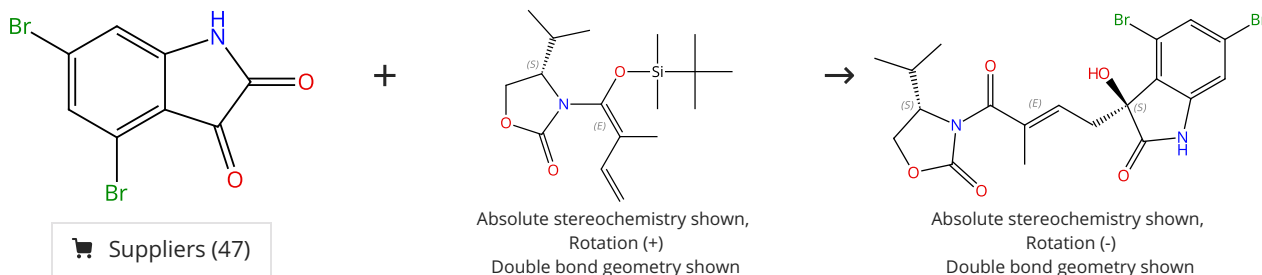
Steps: 1 Yield: 100%



31-330-CAS-3995782	Steps: 1 Yield: 100%	A Catalytic Asymmetric Vinylogous Mukaiyama Aldol Reaction
1.1 Catalysts: Titanium isopropoxide, (-)-Binol Solvents: Diethyl ether, Dichloromethane; 1 h, 40 °C		By: Heumann, Lars V.; et al
1.2 20 min, rt		Organic Letters (2007), 9(21), 4275-4278.
1.3 Reagents: Trimethyl borate; rt		
1.4 0.5 h, -78 °C; -78 °C → rt; 96 h, rt		
Experimental Protocols		

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%

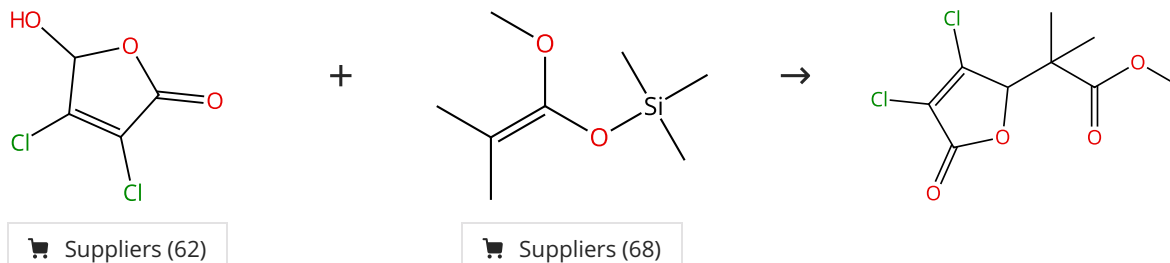


Suppliers (47)

31-330-CAS-7314504	Steps: 1 Yield: 100%	Enantioselective Total Synthesis of Convolutamydines B and E
1.1 Reagents: Titanium tetrachloride Solvents: Dichloromethane; -78 °C; 30 min, 0 °C		By: Nakamura, Tomoaki; et al
1.2 Solvents: Dichloromethane; 6 h, 0 °C		Organic Letters (2006), 8(4), 677-679.
1.3 Reagents: Pyridine		
Experimental Protocols		

Scheme 14 (1 Reaction)

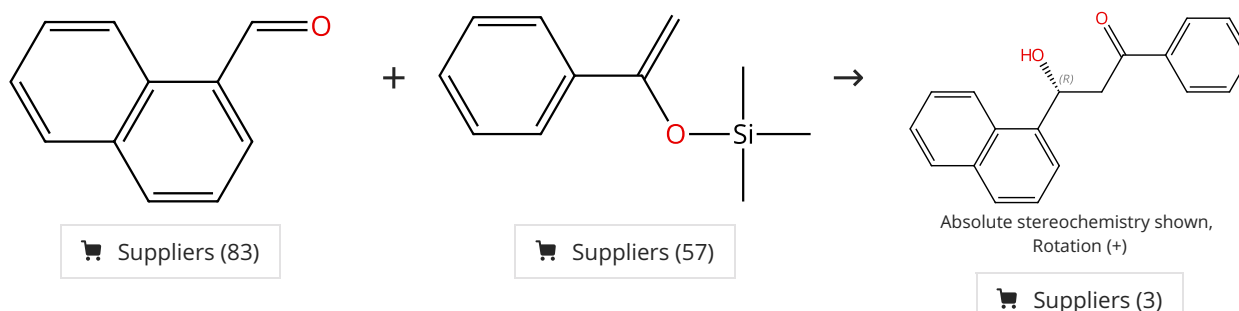
Steps: 1 Yield: 100%



31-330-CAS-3449717	Steps: 1 Yield: 100%	Mucohalic acid in Lewis acid catalyzed Mukaiyama aldol reaction: a concise method for highly functionalized γ-substituted γ-butenolides
1.1 Catalysts: Zinc chloride Solvents: Toluene; rt → -20 °C		By: Angell, Paul; et al
1.2 2 h, -20 °C; 3.5 h, -20 °C → rt		Tetrahedron Letters (2005), 46(12), 2029-2032.

Scheme 15 (1 Reaction)

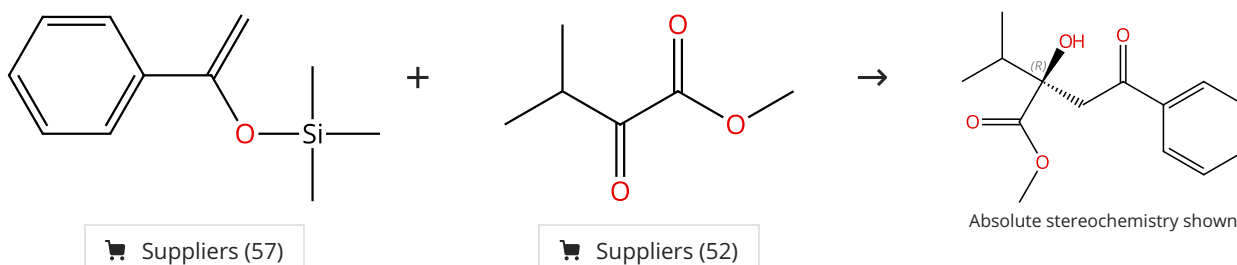
Steps: 1 Yield: 100%



31-330-CAS-4486363 Steps: 1 Yield: 100% 1.1 Catalysts: Silver, (acetato-κO,κO')[(1<i>S</i>)-[1,1'-binaphthalene]-2,2'-diylbis[diphenylphosphine-κP]]-, (7<i>S</i>)- Solvents: Dimethylformamide 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water 1.4 Solvents: Diethyl ether	Mechanism of silver(I)-catalyzed Mukaiyama aldol reaction: active species in solution in AgPF₆-(<i>S</i>)-BINAP versus AgOAc-(<i>S</i>)-BINAP systems By: Ohkouchi, M.; et al Journal of Molecular Catalysis A: Chemical (2001), 170(1-2), 1-15.
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Scheme 16 (1 Reaction)

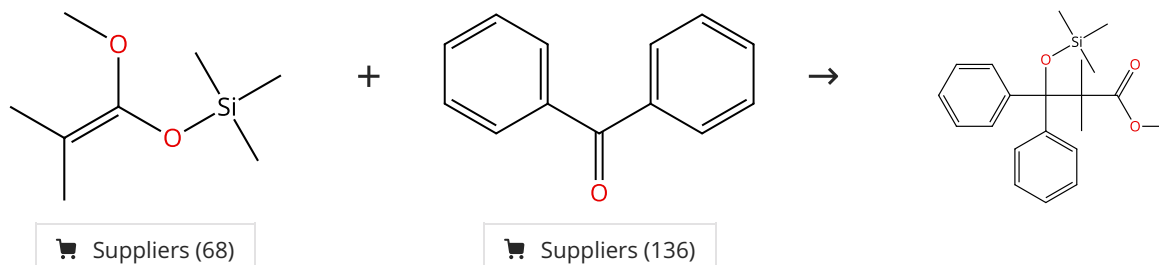
Steps: 1 Yield: 100%



31-330-CAS-13898279 Steps: 1 Yield: 100% 1.1 Catalysts: Copper(II) triflate, (4<i>S</i>)-<i>N</i>-[(4<i>S</i>)-4,5-Dihydro-4-(1-methylethyl)-2-oxazolyl]-4,5-dihydro-<i>N</i>-methyl-4-(1-methylethyl)-2-oxazolamine Solvents: Dichloromethane; 15 min, rt 1.2 Solvents: Dichloromethane; 24 h, rt	Comparison of immobilized Box and azaBox-Cu(II) complexes as catalysts for enantioselective Mukaiyama aldol reactions By: Fraile, Jose M.; et al Journal of Catalysis (2007), 252(2), 303-311.
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Scheme 17 (1 Reaction)

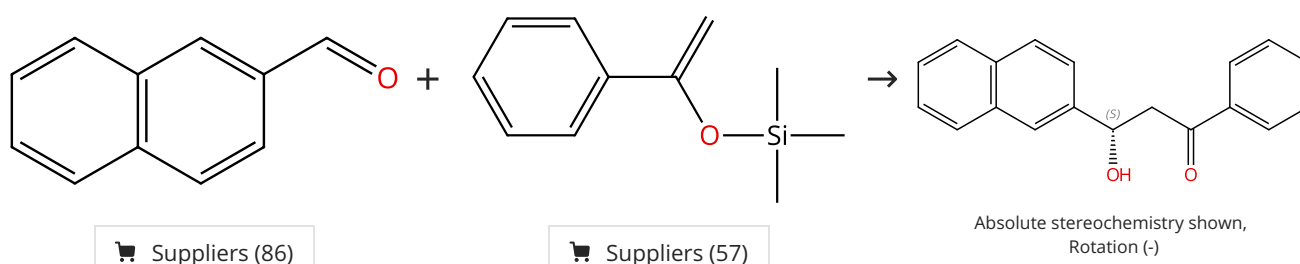
Steps: 1 Yield: 100%



31-330-CAS-14556385 Steps: 1 Yield: 100% 1.1 Reagents: Sodium bis(trimethylsilyl)amide Catalysts: 1,1'-(1,2-Phenylene)bis[1,1-diphenylphosphine oxide], Methyl β-hydroxy-α,α-dimethyl-β-phenylbenzenepanoate Solvents: Tetrahydrofuran; 10 min, rt; 5 min, -78 °C 1.2 Catalysts: Trimethylphenoxysilane; 2 h, -78 °C Experimental Protocols	Sodium Phenoxide-Phosphine Oxides as Extremely Active Lewis Base Catalysts for the Mukaiyama Aldol Reaction with Ketones By: Hatano, Manabu; et al Organic Letters (2007), 9(22), 4527-4530.
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Scheme 18 (2 Reactions)

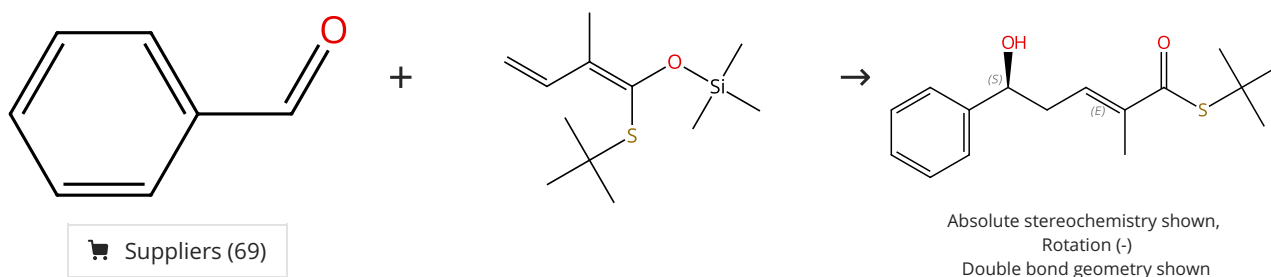
Steps: 1 Yield: 100%



31-330-CAS-13327907 Steps: 1 Yield: 100% 1.1 Catalysts: Silver(1+), [(1<i>S</i>)-[1,1'-binaphthalene]-2,2'-diylbis[diphenylphosphine-κ<i>P</i>]]bis(<i>N,N</i>-dimethylformamide-κ<i>O</i>)-, (<i>T</i>-4)-, hexafluorophosphate(1-) Solvents: Dimethylformamide 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water 1.4 Solvents: Diethyl ether	Mechanism of silver(I)-catalyzed Mukaiyama aldol reaction: active species in solution in AgPF₆-(<i>S</i>)-BINAP versus AgOAc-(<i>S</i>)-BINAP systems By: Ohkouchi, M.; et al Journal of Molecular Catalysis A: Chemical (2001), 170(1-2), 1-15.
31-330-CAS-8516945 Steps: 1 Yield: 100% 1.1 Catalysts: Silver hexafluorophosphate, (-)-BINAP Solvents: Dimethylformamide, Water 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water	Silver(I)-catalyzed asymmetric Mukaiyama aldol reaction By: Ohkouchi, Munetaka; et al Enantiomer (2000), 5(1), 71-81.

Scheme 19 (1 Reaction)

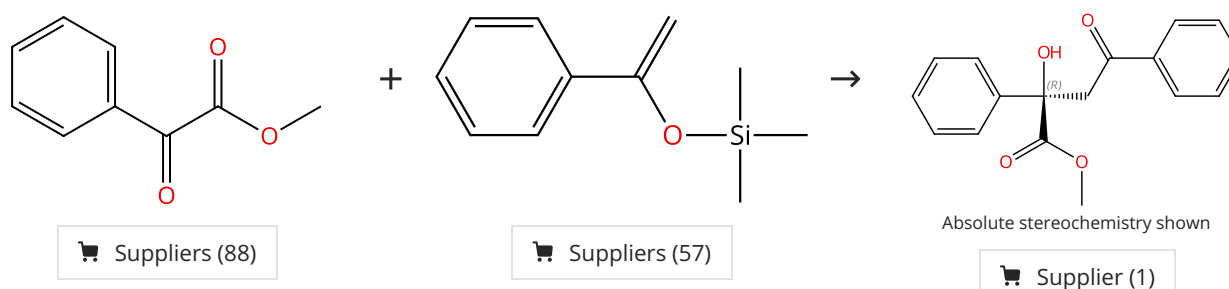
Steps: 1 Yield: 100%



31-330-CAS-1864854 Steps: 1 Yield: 100% 1.1 Catalysts: Titanium isopropoxide, (-)-Binol Solvents: Diethyl ether, Dichloromethane; 1 h, 40 °C 1.2 20 min, rt 1.3 Reagents: Trimethyl borate; rt 1.4 0.5 h, -78 °C; -78 °C → rt; 96 h, rt Experimental Protocols	A Catalytic Asymmetric Vinylogous Mukaiyama Aldol Reaction By: Heumann, Lars V.; et al Organic Letters (2007), 9(21), 4275-4278.
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Scheme 20 (1 Reaction)

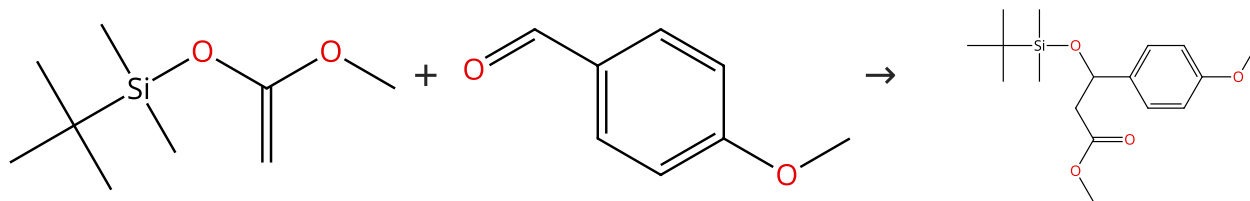
Steps: 1 Yield: 100%



31-330-CAS-6889817 Steps: 1 Yield: 100% 1.1 Catalysts: Copper(II) triflate, (4<i>S</i>)-<i>N</i>-[(4<i>S</i>)-4,5-Dihydro-4-phenyl-2-oxazolyl]-4,5-dihydro-<i>N</i>-methyl-4-phenyl-2-oxazol amine Solvents: Dichloromethane; 15 min, rt 1.2 Solvents: Tetrahydrofuran; 22 h, rt	Comparison of immobilized Box and azaBox-Cu(II) complexes as catalysts for enantioselective Mukaiyama aldol reactions By: Fraile, Jose M.; et al Journal of Catalysis (2007), 252(2), 303-311.
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Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (55)

Suppliers (128)

Supplier (1)

31-330-CAS-1143616

Steps: 1 Yield: 100%

New bis(imino)pyridine complexes of iron(II) and iron(III), and their catalytic activity in the Mukaiyama aldol reaction

By: Shejwalkar, Pushkar; et al

Synthesis (2014), 46(1), 57-66, 10 pp..

1.1 **Catalysts:** Silver hexafluoroantimonate, Dichloro[*N,N'*-[(2,6-pyridinediyl- κ M)diethylidyne]bis[2,6-diethylbenzenamine- κ M]] iron

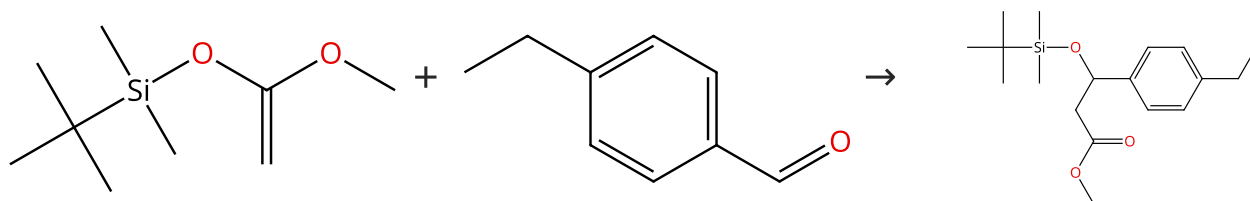
Solvents: Dichloromethane; 1 h, rt

1.2 **Solvents:** Dichloromethane; 3.5 h, rt

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (55)

Suppliers (75)

31-330-CAS-11791893

Steps: 1 Yield: 100%

New bis(imino)pyridine complexes of iron(II) and iron(III), and their catalytic activity in the Mukaiyama aldol reaction

By: Shejwalkar, Pushkar; et al

Synthesis (2014), 46(1), 57-66, 10 pp..

1.1 **Catalysts:** Silver hexafluoroantimonate, Dichloro[*N,N'*-[(2,6-pyridinediyl- κ M)diethylidyne]bis[2,6-diethylbenzenamine- κ M]] iron

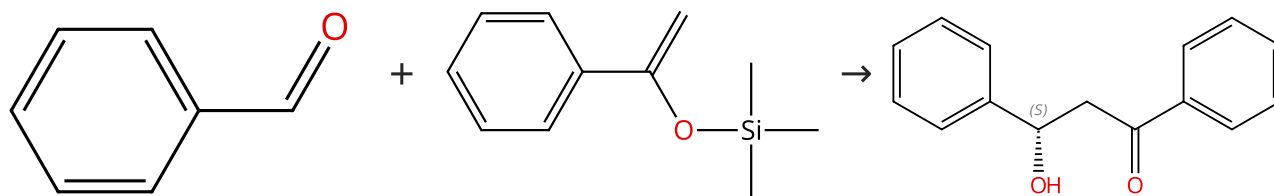
Solvents: Dichloromethane; 1 h, rt

1.2 **Solvents:** Dichloromethane; overnight, rt

Experimental Protocols

Scheme 23 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (69)

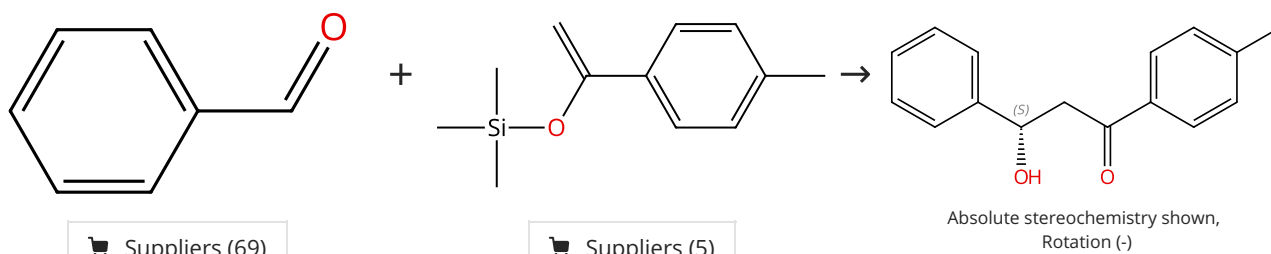
Suppliers (57)

Suppliers (4)

31-330-CAS-5076889 Steps: 1 Yield: 100% 1.1 Catalysts: Silver(1+), [(1<i>S</i>)-[1,1'-binaphthalene]-2,2'-diylbis[diphenylphosphine-κ<i>P</i>]]bis(<i>N,N</i>-dimethylformamide-κ<i>O</i>)-, (<i>T</i>-4)-, hexafluorophosphate(1-) Solvents: Dimethylformamide 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water 1.4 Solvents: Diethyl ether	Mechanism of silver(I)-catalyzed Mukaiyama aldol reaction: active species in solution in AgPF₆-(<i>S</i>)-BINAP versus AgOAc-(<i>S</i>)-BINAP systems By: Ohkouchi, M.; et al Journal of Molecular Catalysis A: Chemical (2001), 170(1-2), 1-15.
31-330-CAS-2195816 Steps: 1 Yield: 100% 1.1 Catalysts: Silver hexafluorophosphate, (-)-BINAP Solvents: Dimethylformamide, Water 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water	Silver(I)-catalyzed asymmetric Mukaiyama aldol reaction By: Ohkouchi, Munetaka; et al Enantiomer (2000), 5(1), 71-81.

Scheme 24 (1 Reaction)

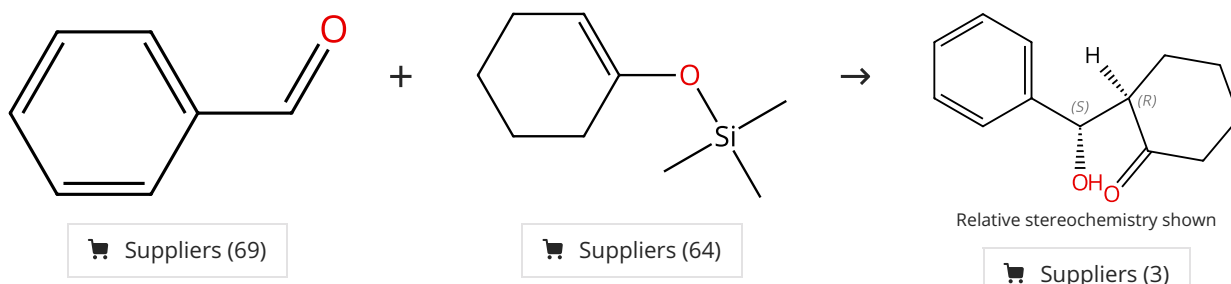
Steps: 1 Yield: 100%



31-330-CAS-15191945 Steps: 1 Yield: 100% 1.1 Catalysts: Silver hexafluorophosphate, (-)-BINAP Solvents: Dimethylformamide, Water 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water	Silver(I)-catalyzed asymmetric Mukaiyama aldol reaction By: Ohkouchi, Munetaka; et al Enantiomer (2000), 5(1), 71-81.
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Scheme 25 (1 Reaction)

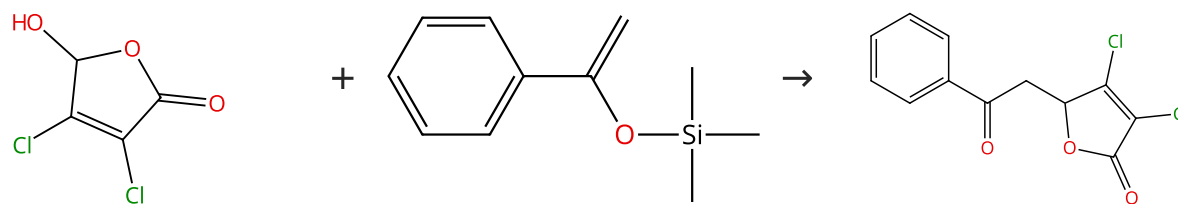
Steps: 1 Yield: 100%



31-330-CAS-8188031 Steps: 1 Yield: 100% 1.1 Catalysts: Silver acetate, (-)-BINAP Solvents: Dimethylformamide, Water 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water	Silver(I)-catalyzed asymmetric Mukaiyama aldol reaction By: Ohkouchi, Munetaka; et al Enantiomer (2000), 5(1), 71-81.
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Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (62)

Suppliers (57)

Suppliers (7)

31-330-CAS-14118232

Steps: 1 Yield: 100%

Mucohalic acid in Lewis acid catalyzed Mukaiyama aldol reaction: a concise method for highly functionalized γ -substituted γ -butenolides

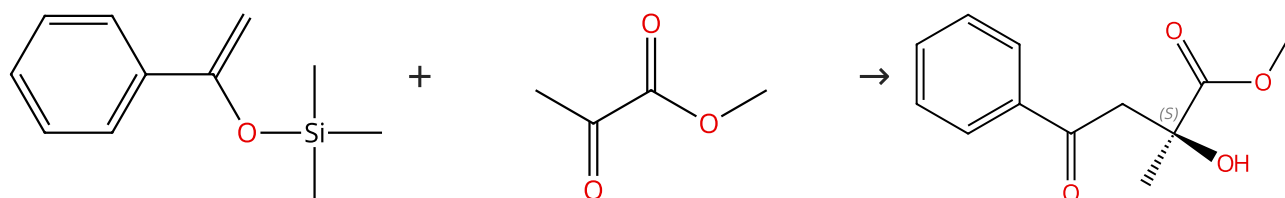
By: Angell, Paul; et al

Tetrahedron Letters (2005), 46(12), 2029-2032.

1.1 **Catalysts:** Scandium triflate**Solvents:** Diethyl ether; rt $\rightarrow$ -20 °C1.2 1 h, -20 °C; 3 h, -20 °C $\rightarrow$ 15 °C; 16 h, 15 °C

Scheme 27 (5 Reactions)

Steps: 1 Yield: 100%



Suppliers (57)

Suppliers (64)

Supplier (1)

31-330-CAS-5710437

Steps: 1 Yield: 100%

Comparison of immobilized Box and azaBox-Cu(II) complexes as catalysts for enantioselective Mukaiyama aldol reactions

By: Fraile, Jose M.; et al

Journal of Catalysis (2007), 252(2), 303-311.

1.1 **Catalysts:** Copper(II) triflate, (4*S*)-4-(1,1-Dimethylethyl)-*N*-[(4*S*)-4-(1,1-dimethylethyl)-4,5-dihydro-2-oxazolyl]-4,5-dihydro-*N*-methyl-2-oxazoline**Solvents:** Dichloromethane; 15 min, rt1.2 **Solvents:** Tetrahydrofuran; 24 h, rt

31-330-CAS-5063167

Steps: 1 Yield: 100%

Comparison of immobilized Box and azaBox-Cu(II) complexes as catalysts for enantioselective Mukaiyama aldol reactions

By: Fraile, Jose M.; et al

Journal of Catalysis (2007), 252(2), 303-311.

1.1 **Catalysts:** (*SP*4-2)-[(4*S*,4'*S*)-2,2'-(1-Methylethylidene)bis[4,5-dihydro-4-phenyloxazole- κ ^{N3}]]bis(1,1,1-trifluoromethanesulfonate- κ ^O)copper (laponite-supported)**Solvents:** Dichloromethane; 24 h, rt

31-330-CAS-2744586

Steps: 1 Yield: 100%

Multipurpose box- and azabox-based immobilized chiral catalysts

By: Fraile, Jose M.; et al

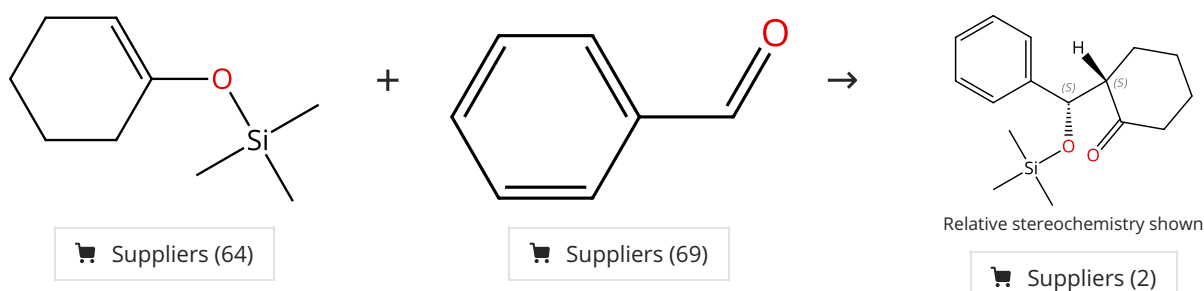
Advanced Synthesis & Catalysis (2006), 348(12+13), 1680-1688.

1.1 **Catalysts:** Copper(II) triflate, (*S,S*)-2,2'-Isopropylidenebis[4-phenyl-2-oxazoline]**Solvents:** Dichloromethane; 15 min, rt1.2 **Catalysts:** Lithium magnesium sodium silicate**Solvents:** Methanol; 24 h, rt1.3 **Solvents:** Dichloromethane; 48 h, rt

31-330-CAS-8461941 Steps: 1 Yield: 100% 1.1 Catalysts: Copper(II) triflate, (<i>S,S</i>)-2,2'-Isopropylidenebis[4-phenyl-2-oxazoline] Solvents: Dichloromethane; 15 min, rt 1.2 Catalysts: Nafion 50 Solvents: Methanol; 24 h, rt 1.3 Solvents: Dichloromethane; 96 h, rt	Multipurpose box- and azabox-based immobilized chiral catalysts By: Fraile, Jose M.; et al Advanced Synthesis & Catalysis (2006), 348(12+13), 1680-1688.
31-330-CAS-6611134 Steps: 1 Yield: 100% 1.1 Catalysts: Copper(II) triflate, (4 <i>S</i>)-4-(1,1-Dimethylethyl)- <i>N</i> -[(4 <i>S</i>)-4-(1,1-dimethylethyl)-4,5-dihydro-2-oxazolyl]-4,5-dihydro- <i>N</i> -methyl-2-oxazoline Solvents: Dichloromethane; 15 min, rt 1.2 Solvents: Tetrahydrofuran; 24 h, rt	Multipurpose box- and azabox-based immobilized chiral catalysts By: Fraile, Jose M.; et al Advanced Synthesis & Catalysis (2006), 348(12+13), 1680-1688.

Scheme 28 (1 Reaction)

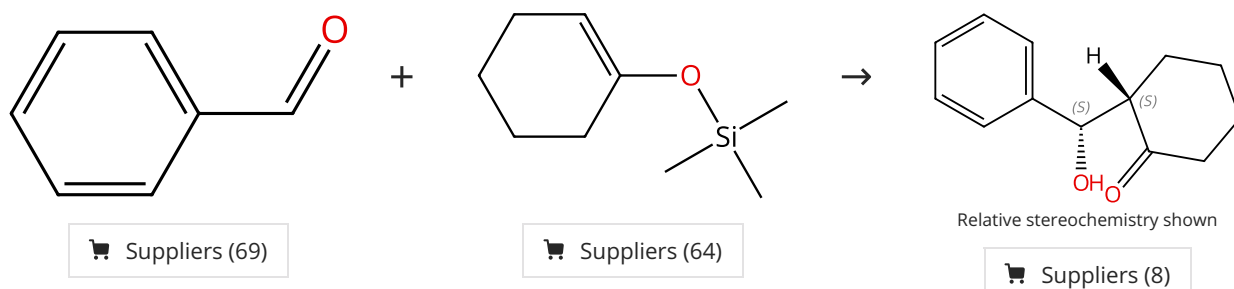
Steps: 1 Yield: 100%



31-330-CAS-10808028 Steps: 1 Yield: 100% 1.1 Catalysts: Scandium triflate Solvents: Dichloromethane; 15 min, 20 °C Experimental Protocols	Sc^{III}-Doped Zeolites as New Heterogeneous Catalysts: Mukaiyama Aldol Reaction By: Olmos, Andrea; et al Chemistry - A European Journal (2009), 15(42), 11229-11234, S11229/1-S11229/11.
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Scheme 29 (1 Reaction)

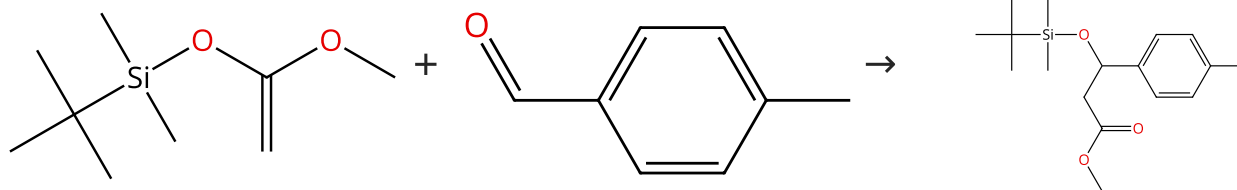
Steps: 1 Yield: 100%



31-330-CAS-2026180 Steps: 1 Yield: 100% 1.1 Catalysts: Silver, (acetato-κ ^O ,κ ^O)[(1 <i>S</i>)-[1,1'-binaphthalene]-2,2'-diylbis[diphenylphosphine-κ ^P]-, (7-4)- Solvents: Dimethylformamide 1.2 - 1.3 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water 1.4 Solvents: Diethyl ether	Mechanism of silver(I)-catalyzed Mukaiyama aldol reaction: active species in solution in AgPF₆-(<i>S</i>)-BINAP versus AgOAc-(<i>S</i>)-BINAP systems By: Ohkouchi, M.; et al Journal of Molecular Catalysis A: Chemical (2001), 170(1-2), 1-15.
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Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (55)

Suppliers (104)

Supplier (1)

31-330-CAS-9998869

Steps: 1 Yield: 100%

New bis(imino)pyridine complexes of iron(II) and iron(III), and their catalytic activity in the Mukaiyama aldol reaction

By: Shejwalkar, Pushkar; et al

Synthesis (2014), 46(1), 57-66, 10 pp..

1.1 **Catalysts:** Silver hexafluoroantimonate, Dichloro[*N,N'*-(2,6-pyridinediyl-κ*M*)diethylidyne]bis[2,6-diethylbenzamine-κ*M*] iron

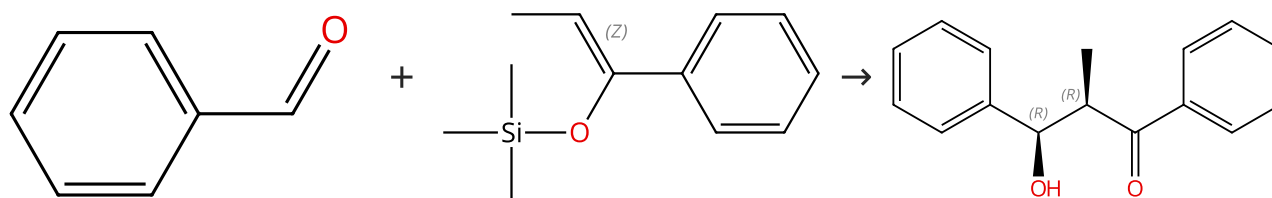
Solvents: Dichloromethane; 1 h, rt

1.2 **Solvents:** Dichloromethane; 3.5 h, rt

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

Double bond geometry shown

Suppliers (3)

Relative stereochemistry shown

Supplier (1)

31-330-CAS-6063343

Steps: 1 Yield: 100%

Silver(I)-catalyzed asymmetric Mukaiyama aldol reaction

By: Ohkouchi, Munetaka; et al

Enantiomer (2000), 5(1), 71-81.

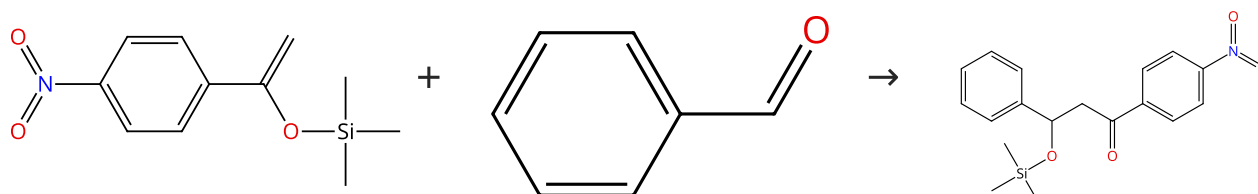
1.1 **Catalysts:** Silver acetate, (-)-BINAP
Solvents: Dimethylformamide, Water

1.2 -

1.3 **Reagents:** Hydrochloric acid
Solvents: Tetrahydrofuran, Water

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

Suppliers (69)

Supplier (1)

31-330-CAS-10909559

Steps: 1 Yield: 100%

Sc^{III}-Doped Zeolites as New Heterogeneous Catalysts: Mukaiyama Aldol Reaction

By: Olmos, Andrea; et al

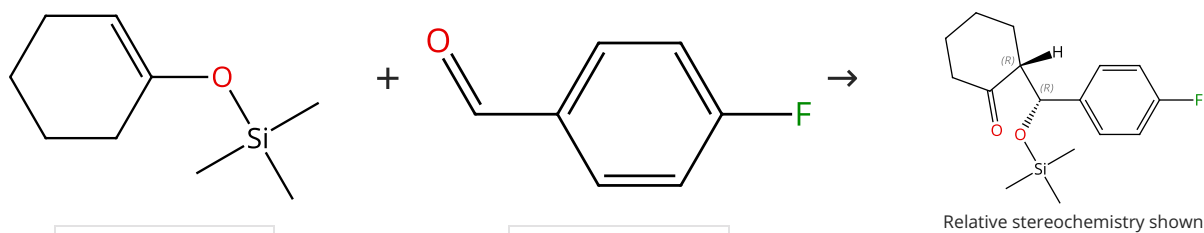
Chemistry - A European Journal (2009), 15(42), 11229-11234, S11229/1-S11229/11.

1.1 **Catalysts:** Scandium triflate
Solvents: Dichloromethane; 15 min, 20 °C

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (64)

Suppliers (96)

31-330-CAS-3894690

Steps: 1 Yield: 100%

Sc^{III}-Doped Zeolites as New Heterogeneous Catalysts: Mukaiyama Aldol Reaction

By: Olmos, Andrea; et al

Chemistry - A European Journal (2009), 15(42), 11229-11234, S11229/1-S11229/11.

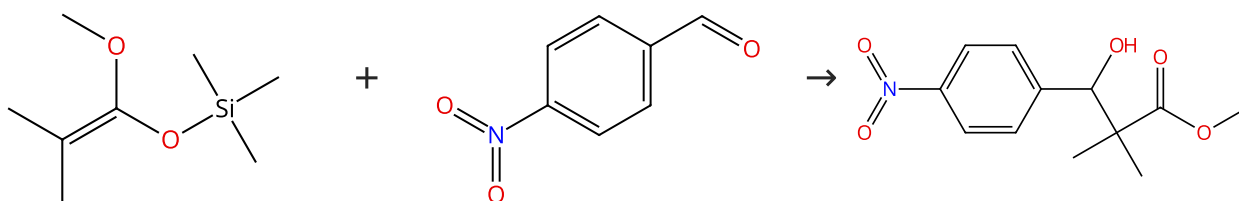
1.1 Catalysts: Scandium triflate

Solvents: Dichloromethane; 15 min, 20 °C

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

Suppliers (92)

Suppliers (2)

31-330-CAS-6005693

Steps: 1 Yield: 100%

A versatile catalyst for asymmetric reactions of carbonyl groups working purely by activation through hydrogen bonding: Mukaiyama-aldol, hetero Diels-Alder and Friedel-Crafts reactions

By: Zhuang, Wei; et al

Organic & Biomolecular Chemistry (2005), 3(18), 3284-3289.

1.1 Catalysts: *N,N'*-[(1*R*,2*R*)-1,2-Diphenyl-1,2-ethanediyl]bis[1,1,1-trifluoromethanesulfonamide]

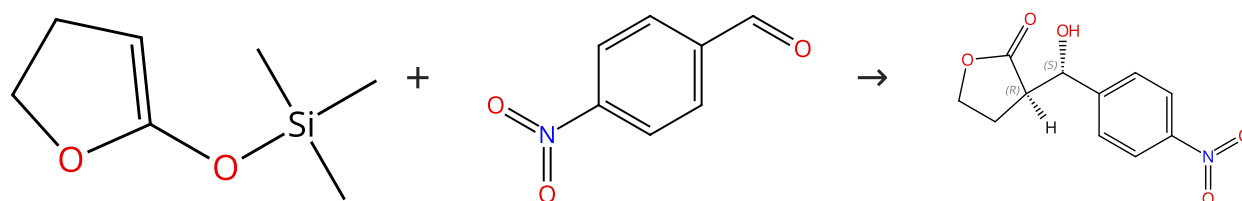
Solvents: Chloroform; 40 h, -50 °C

1.2 Reagents: Trifluoroacetic acid

Solvents: Dichloromethane; 1 h, rt

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (5)

Suppliers (92)

31-330-CAS-4476798

Steps: 1 Yield: 100%

Oxovanadium(IV) complex-mediated catalyst in Mukaiyama-type aldol additions

By: Chen, Chien-Tien; et al

Synlett (1999), (6), 816-818.

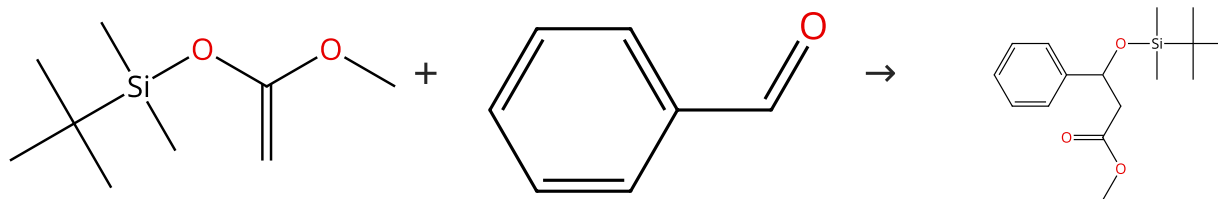
1.1 Catalysts: [[1,1'-Biphenyl]-2,2'-diolato(2-)-κO,κO']bis(1-methyl-1*H*-imidazole-κ*N*³)oxovanadium

Solvents: Dichloromethane

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (55)

Suppliers (69)

Suppliers (2)

31-330-CAS-10488740

Steps: 1 Yield: 100%

Carbon-carbon bond formation catalyzed by lithium perchlorate in dichloromethane

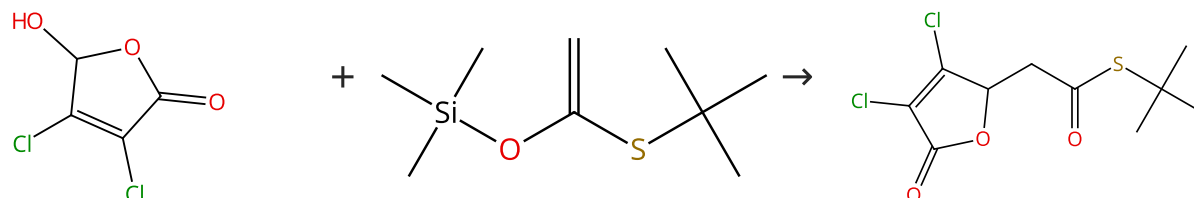
1.1 **Catalysts:** Lithium perchlorate
Solvents: Dichloromethane

By: Reetz, Manfred T.; et al

Tetrahedron Letters (1993), 34(7), 1119-22.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (62)

Suppliers (7)

31-330-CAS-11988285

Steps: 1 Yield: 100%

Mucohalic acid in Lewis acid catalyzed Mukaiyama aldol reaction: a concise method for highly functionalized γ -substituted γ -butenolides

1.1 **Catalysts:** Zinc chloride
Solvents: Toluene; rt $\rightarrow$ -20 °C

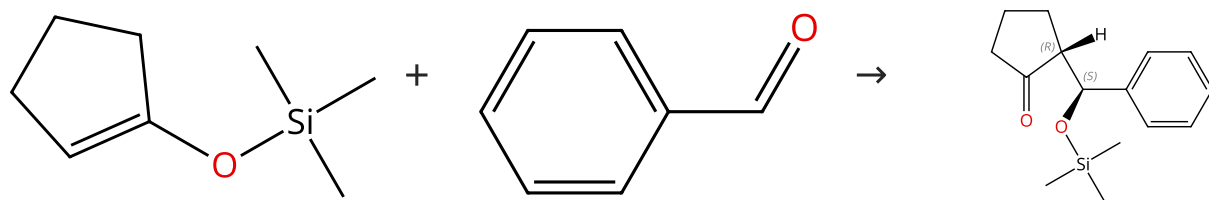
By: Angell, Paul; et al

1.2 2 h, -20 °C; 3.5 h, -20 °C $\rightarrow$ rt; 3 d, rt

Tetrahedron Letters (2005), 46(12), 2029-2032.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (59)

Suppliers (69)

Relative stereochemistry shown

Supplier (1)

31-330-CAS-8495803

Steps: 1 Yield: 100%

Sc^{III}-Doped Zeolites as New Heterogeneous Catalysts: Mukaiyama Aldol Reaction

1.1 **Catalysts:** Scandium triflate
Solvents: Dichloromethane; 15 h, 20 °C

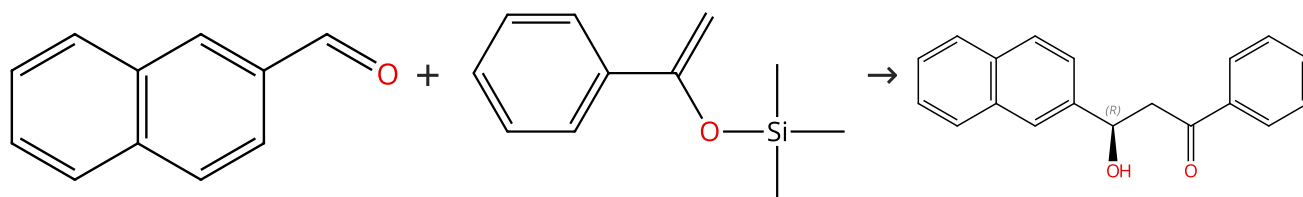
By: Olmos, Andrea; et al

Experimental Protocols

Chemistry - A European Journal (2009), 15(42), 11229-11234, S11229/1-S11229/11.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (57)

Absolute stereochemistry shown,
Rotation (+)

Supplier (1)

31-330-CAS-8743906

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Silver, (acetato- $\kappa O, \kappa O'$)[(1*S*)-[1,1'-binaphthalene]-2,2'-diylbis[diphenylphosphine- κP]-, (*T*-4)-
Solvents: Dimethylformamide
- 1.2 -
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Tetrahydrofuran, Water
- 1.4 **Solvents:** Diethyl ether

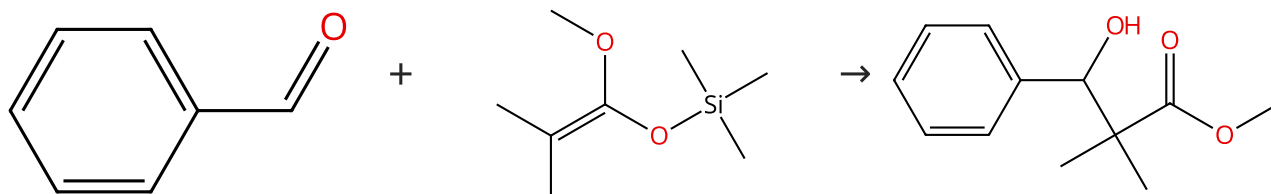
Mechanism of silver(I)-catalyzed Mukaiyama aldol reaction: active species in solution in AgPF₆-(*S*)-BINAP versus AgOAc-(*S*)-BINAP systems

By: Ohkouchi, M.; et al

Journal of Molecular Catalysis A: Chemical (2001), 170(1-2), 1-15.

Scheme 40 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (69)

Suppliers (68)

Suppliers (11)

31-330-CAS-9027252

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Silver, (acetato- $\kappa O, \kappa O'$)[(1*S*)-[1,1'-binaphthalene]-2,2'-diylbis[diphenylphosphine- κP]-, (*T*-4)-
Solvents: Dimethylformamide
- 1.2 -
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Tetrahydrofuran, Water
- 1.4 **Solvents:** Diethyl ether

Mechanism of silver(I)-catalyzed Mukaiyama aldol reaction: active species in solution in AgPF₆-(*S*)-BINAP versus AgOAc-(*S*)-BINAP systems

By: Ohkouchi, M.; et al

Journal of Molecular Catalysis A: Chemical (2001), 170(1-2), 1-15.

31-330-CAS-10319762

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Silver acetate, (-)-BINAP
Solvents: Dimethylformamide, Water
- 1.2 -
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Tetrahydrofuran, Water

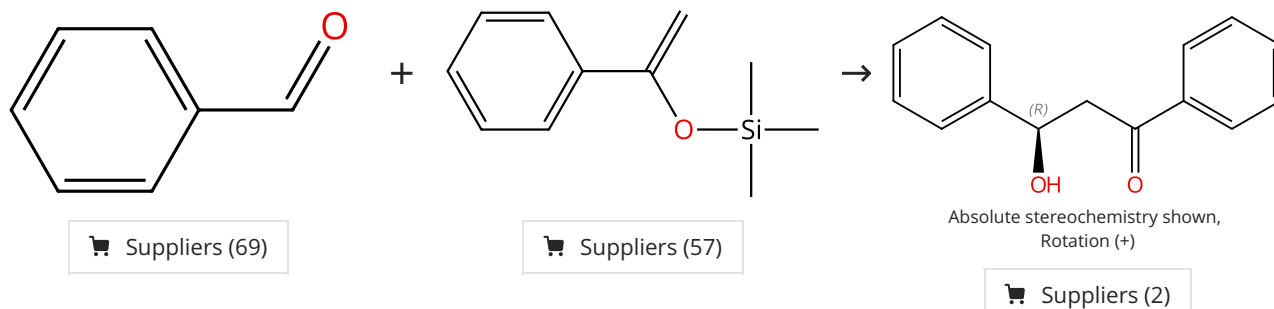
Silver(I)-catalyzed asymmetric Mukaiyama aldol reaction

By: Ohkouchi, Munetaka; et al

Enantiomer (2000), 5(1), 71-81.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



31-330-CAS-17989

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Silver, (acetato-κO,κO')[(1*S*)-[1,1'-binaphthalene]-2,2'-diylbis[diphenylphosphine-κP]]-, (*T*-4)-
Solvents: Dimethylformamide
- 1.2 -
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Tetrahydrofuran, Water
- 1.4 **Solvents:** Diethyl ether

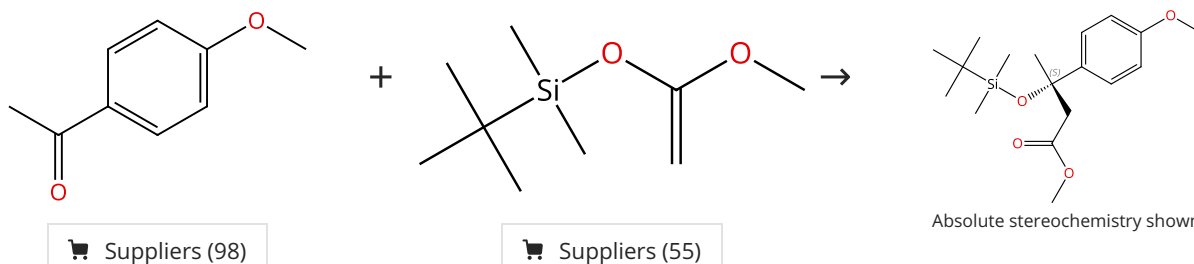
Mechanism of silver(I)-catalyzed Mukaiyama aldol reaction: active species in solution in AgPF₆-(*S*)-BINAP versus AgOAc-(*S*)-BINAP systems

By: Ohkouchi, M.; et al

Journal of Molecular Catalysis A: Chemical (2001), 170(1-2), 1-15.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 99%



31-330-CAS-22174263

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** 2241514-78-1
Solvents: Diethyl ether; 10 min, -78 °C
- 1.2 -78 °C; -78 °C → -20 °C; 12 h, -20 °C

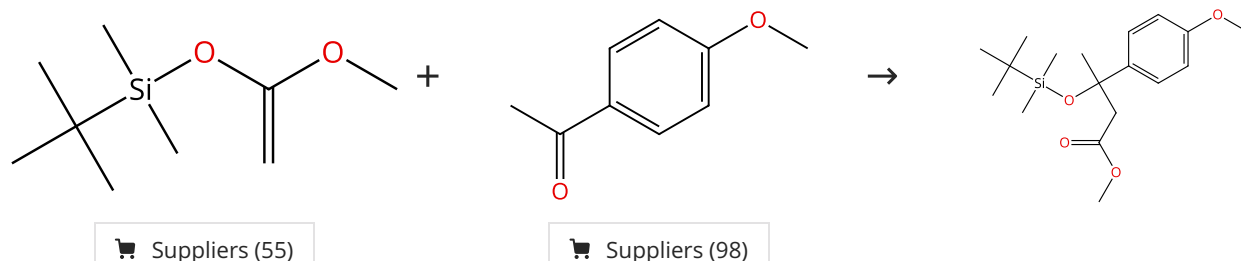
Approaching sub-ppm-level asymmetric organocatalysis of a highly challenging and scalable carbon-carbon bond forming reaction

By: Bae, Han Yong; et al

Nature Chemistry (2018), 10(8), 888-894.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%



31-330-CAS-22169660

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** Bis(trifluoromethanesulfonyl)imide
Solvents: Diethyl ether; -78 °C; -78 °C → -20 °C; 30 min, -20 °C
- Experimental Protocols

Triflimide: An Overlooked High-Performance Catalyst of the Mukaiyama Aldol Reaction of Silyl Ketene Acetals with Ketones

By: Bae, Han Yong; et al

Chemistry - A European Journal (2018), 24(52), 13767-13772.

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